ASSIGNMENT-1

NAME:-NITHISH

BATCH-16

HTNO:-2303A51048

Task 1: AI-Generated Logic Without Modularisation *(Factorial without Functions)*

Thonny - <untitled> @ 7 : 23

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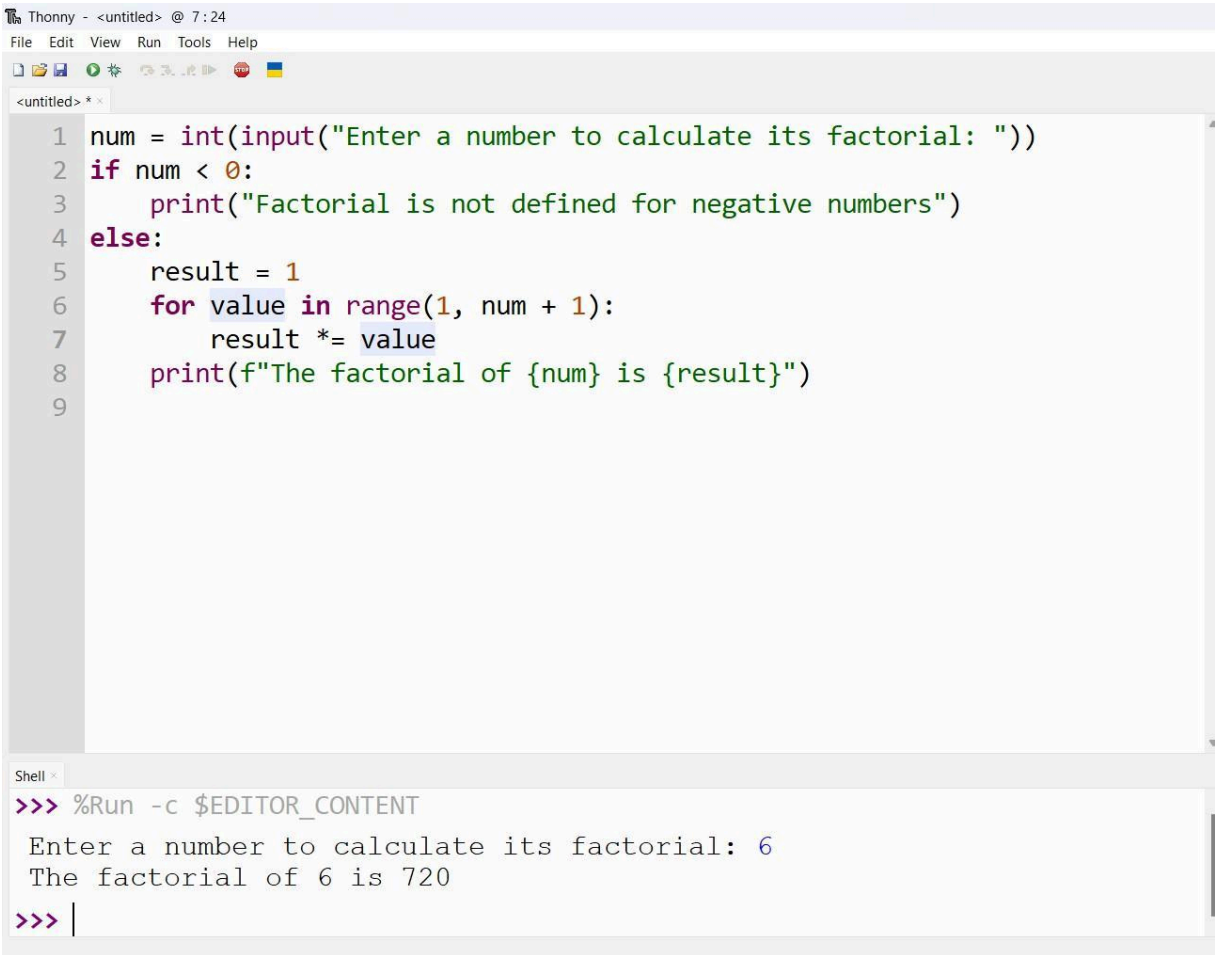
```python
num = int(input("Enter a number to calculate its factorial: "))
if num < 0:
    print("Factorial is not defined for negative numbers")
else:
    factorial = 1
    for i in range(1, num + 1):
        factorial *= i
    print(f"The factorial of {num} is {factorial}")
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Enter a number to calculate its factorial: 5
 The factorial of 5 is 120
>>>
```

Task 2: AI Code Optimization & Cleanup (Improving Efficiency and Readability)



Task 3: Modular Design Using AI Assistance (Factorial with Functions)

Thonny - <untitled> @ 9 : 34

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```python
def calculate_factorial(n):
    if n < 0:
        return "Factorial is not defined for negative numbers"
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result
num = int(input("Enter a number to calculate its factorial: "))
output = calculate_factorial(num)
if isinstance(output, str):
    print(output)
else:
    print(f"The factorial of {num} is {output}")
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Enter a number to calculate its factorial: 5
 The factorial of 5 is 120
>>>
```

Task 4: Comparative Analysis – Procedural vs Modular AI Code

Procedural (Without Function):-

Thonny - <untitled> @ 11 : 1

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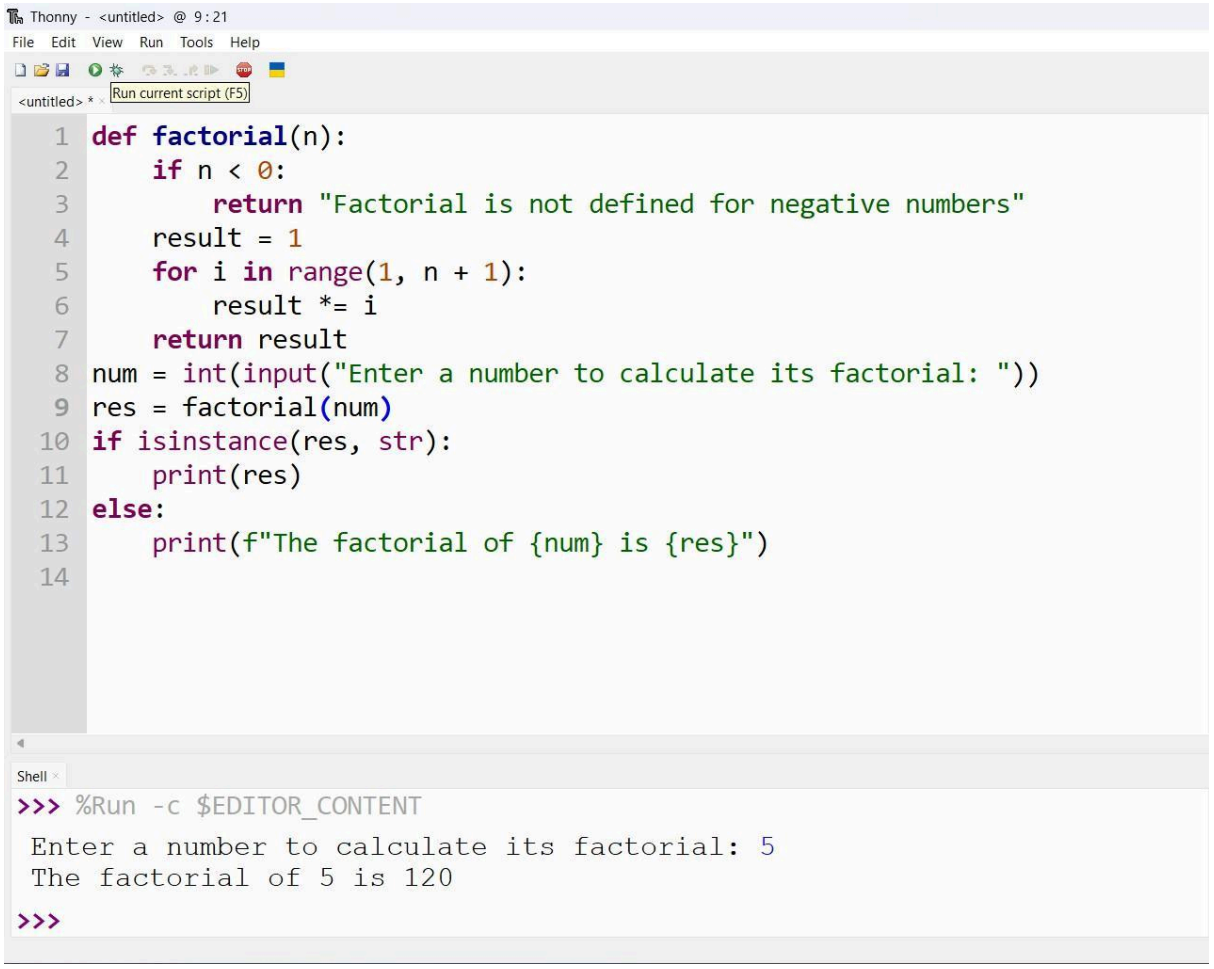
```python
num = int(input("Enter a number to calculate its factorial: "))

if num < 0:
    print("Factorial is not defined for negative numbers")
else:
    result = 1
    for i in range(1, num + 1):
        result *= i

    print(f"The factorial of {num} is {result}")
```

Shell

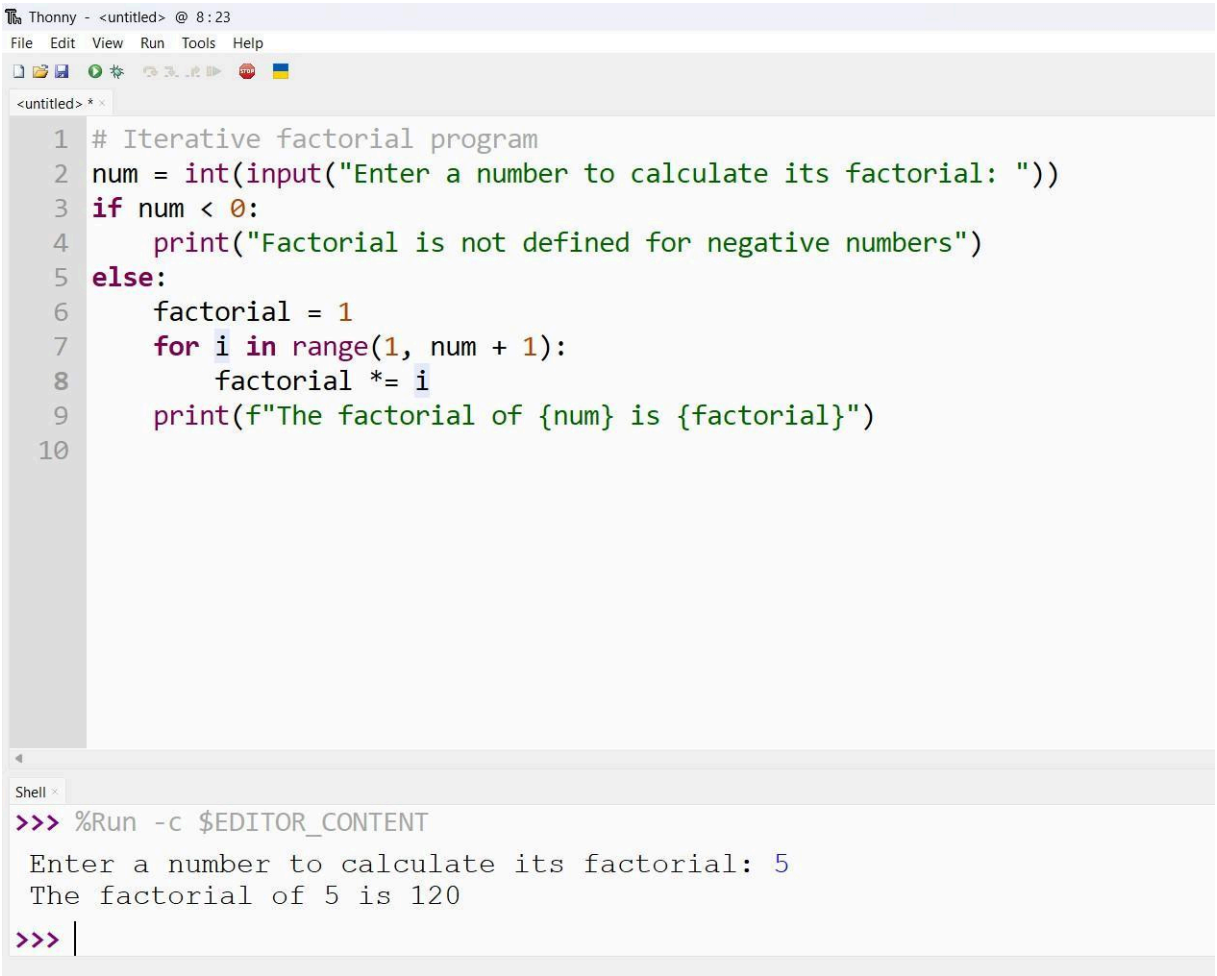
```
>>> %Run -c $EDITOR_CONTENT
 Enter a number to calculate its factorial: 4
 The factorial of 4 is 24
>>>
```

Modular (With Function)



Task 5: AI-Generated Iterative vs Recursive Thinking

Iterative Approach



Recursive Approach

Thonny - <untitled> @ 1 : 30

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```python
# Recursive factorial program
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)
num = int(input("Enter a number to calculate its factorial: "))
if num < 0:
    print("Factorial is not defined for negative numbers")
else:
    result = factorial(num)
    print(f"The factorial of {num} is {result}")
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Enter a number to calculate its factorial: 5
 The factorial of 5 is 120
>>>
```